(17) $(g \circ f)(x) = 62.115x$

(18) day revenue = $(200 - x)x$; total cost =$100(200 - x)$;

total profit $= (200 - x)x - 100(200 - x) in rupees$

(19) $f^{-1}(x) = \frac{9x}{5} + 32$ (20) $f^{-1}(x) = \frac{x+4}{3}$

Exercise 1.5

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
(3)	(2)	(4)	(1)	(1)	(4)	(2)	(2)	(3)	(2)	(2)	(3)	(3)	(2)	(4)
16	17	18	19	20	21	22	23	24	25					
(3)	(4)	(3)	(2)	(4)	(4)	(1)	(4)	(2)	(3)					

Exercise 2.1

(1) $\sqrt{7} \in \mathbb{R} - \mathbb{Q}$, $-\frac{1}{4} \in \mathbb{Q}$, $0 \in \mathbb{Z}, \mathbb{Q}$,

$3.14 \in \mathbb{Q}$, $4 \in \mathbb{N}, \mathbb{Z}, \mathbb{Q}$, $\frac{22}{7} \in \mathbb{Q}$

(3) yes, $4 + \sqrt{3}$, $2 + \sqrt{3}$ (4) yes, $2 + \sqrt{3}, 2 - \sqrt{3}$ (5) $\frac{1}{2^{1001}}$

Exercise 2.2

(1)(i) $-4 < x < 10$ (ii) $\mathbb{R}$ (iii) $\frac{11}{3} \leq x \leq \frac{13}{3}$ (iv) $-7 < x < 7$

(2) $\left(-\infty, \frac{5}{12}\right) \cup \left(\frac{7}{12}, \infty\right)$ (3) $\left(-\infty, \frac{-7}{3}\right] \cup \left[\frac{7}{3}, \infty\right)$ (4) $-\frac{15}{2} \leq x \leq \frac{11}{2}$

(5) $-\frac{3}{10} < x < \frac{7}{10}$ (6) no solution

Exercise 2.3

(1)(i) $[-1, 4)$ (ii) $[-3, 5]$ (iii) $(-\infty, 3)$ (iv) $(-\infty, 5)$

(2)(i) $1, 2, 3, 4$ (ii) $\ldots, -3, -2, -1, 0, 1, 2, 3, 4$

(3)(i) $\left(-\infty, -\frac{9}{2}\right]$ (ii) $\ldots, -7, -6, -5$ (iii) no solution (4) (i) (-∞, 2] (ii) $\left(\frac{34}{5}, \infty\right)$

(5) 93 (6) between120ℓ and 300ℓ (7) $(11, 13), (13, 15), (15, 17), (17, 19)$

(8) $t = 9$seconds, 11seconds (9) less than10 hours (10) less than ₹$21,000$ or greater than ₹33000

Exercise 2.4

(1) $x^2 - 4x - 21 = 0$ (2) $-\frac{2}{5}(x^2 - 2x - 4)$ (3) $x^2 + \frac{\sqrt{2}}{3}x + \frac{1}{3} = 0$

(6) (i) $b = 0$ (ii) $3b^2 = 16ac$ (iii) $c = a$